AMITY UNIVERSITY
PATNA

Course Curriculum

Course Title: Data Science – An Introduction
Course Code: CSIT379
Credit Units: 05
Course Level: UG
Syllabus Version – I

L	T	P/S	SW	No. of PSDA	Total Credit Unit
3	0	4	0	0	5

Course Description:

This course provides an introduction to the core principles, tools, and techniques of data science and analytics. Students will learn how to collect, clean, analyze, and visualize data using industry-standard tools and programming languages such as Python or R. Key topics include exploratory data analysis, statistical inference, data wrangling, machine learning fundamentals, and effective data communication. Emphasis is placed on real-world applications, critical thinking, and hands-on problem-solving with datasets from various domains. By the end of the course, students will be able to derive insights from data and make informed, data-driven decisions.

Course Objectives:

CO1: Define data science and analytics, their importance, and their role in solving real-world problems across various domains.
CO2: Describe and apply the major phases of the data science process, including data collection, cleaning, exploration, modeling, and communication of results.
CO3: Demonstrate the ability to gather data from various sources and preprocess it using techniques such as data cleaning, normalization, and transformation.
CO4: Use descriptive statistics and visualization tools (e.g., histograms, box plots, scatter plots) to understand the structure and patterns in data.
CO5:Utilize programming languages (such as Python or R) and libraries (like Pandas, NumPy, or ggplot2) for basic data manipulation and analysis tasks.

Prerequisites:
Basic knowledge of statistics and programming is recommended.

Course Contents/Syllabus:

	Weightage (%)	Lecture Hour
Module I:	**25**	**20**
Introduction to Data Science, Big Data and Data Science, Introduction to Big Data Platform, Challenges of Conventional Systems, Nature of Data, Analytic Processes and Tools, Analysis vs Reporting, Modern Data Analytic Tools, overview of Power BI and Tableau. Multi-Dimensional data, Basic tools (plots, graphs and summary statistics) of EDA, Needs Preprocessing the Data, Data Cleaning, Data Integration and Transformation, Data Reduction, Discretization and Concept Hierarchy Generation. Data summarization, Data Normalization.		
Module II:	**15**	**10**
Computer Vision, Time series analysis and Forecasting, Streaming Data, Statistical Inference - Populations and samples - Statistical modeling, probability distributions, fitting a model, Feature Generation (role of domain expertise) and Feature Selection algorithms.		

Module III: Data preprocessing, Data Mining Techniques: Statistical techniques, Characterization and discrimination, Association and market basket analysis, Classification and Prediction, Decision trees, Neural Networks, Bayesian Classification, Association rules, Apriori, FP Tree, Introduction to Genetic Algorithm, Cluster analysis, Automatic Cluster Detection, Outlier analysis.	**15**	**15**
Module IV: Raw Text, Sourcing, and Normalization, Tokenization, Lemmatization, Chunking, Sentence Parse, and Dependencies, Information Extraction and Text Classification, Information Extraction and Text Classification, Sentiment Analysis, Recurrent Neural Networks (LSTM).	**15**	**10**
Module V: Decision Trees; Random Forests, Machine Learning Algorithms - Linear Regression, k-Nearest Neighbors (k-NN), k-means, Naive Bayes, SVM, Clustering, Expectation Maximization, Dimensionality Reduction, Feature Selection, PCA.	**15**	**10**
Module VI: Develop a Data Science project under the Viksit Bharat@2047 theme focusing on leveraging data to drive socio-economic progress, improve governance, and enhance the quality of life for citizens.	**15**	**10**

Course Learning Outcomes (CLOs):

CLO1: Understand: Explain the fundamental concepts of data science and analytics, including the data science lifecycle and various types of analytics (descriptive, predictive and prescriptive).
CLO2: Apply: Use appropriate statistical methods and mathematical models to analyze and interpret data sets.
CLO3: Analyze: Demonstrate the ability to collect, preprocess, clean, and manage structured and unstructured data using tools like Python, R, SQL or Excel.
CLO4: Create: Create meaningful data visualizations to communicate findings clearly to stakeholders.

Pedagogy

This course will be taught in blended mode in form of lectures, self-work and covering different modules of the course.

Assessment/ Examination Scheme:

Theory L/T (%)	**Lab/Practical/Studio (%)**	**TOTAL**
60	**40**	**100**

Theory Assessment (L&T):

Continuous Assessment/Internal Assessment (40 %)					End Term Examination (60%)
Components (Drop down)	Viva	Class Test	Home Assignment	Attendance	End Term
Weightage (%)	5	20	10	5	60

Lab/ Practical/ Studio Assessment:

Continuous Assessment/Internal Assessment (40 %)						End Term Examination (60%)	
Components (Drop down)	Class Test (Practical based)	Lab Record	Continuous Performance	Internal Viva	Attendance	Practical Evaluation	Viva
Weightage (%)	10	10	10	5	5	40	20

Introduction to Data Science and Analytics: Lab Problems [60 Hr.]

S.No.	Experiment	Practical Hours
1	To create Line chart, Bar chart and Histogram.	5
2	Create a n * k matrix to represent a linear function that maps k-dimensional vectors to n-dimensional vectors.	5
3	Pandas Word Analysis Word Generation Morphology N-Grams N-Grams Smoothing POS Tagging:	5
4	Hidden Markov Model POS Tagging	5
5	Viterbi Decoding Building POS Tagger Chunking and Building Chunker Sentiment analysis Manipulating and Rescaling data.	5
6	Finding k-nearest neighbor Spam Filter Entropy of Partition.	5
7	Implement an application that stores big data in Hbase / MongoDB / Pig Using Hadoop / R.	5
8	Visualize data using any plotting framework	5
9	Reading Excel data sheet in R	5
10	To get the input from user and perform numerical operations (MAX, MIN, AVG, SUM, SQRT, ROUND) using in R	5
11	To perform data import/export (.CSV, .XLS, .TXT) operations using data frames in R	10

Mapping Continuous Evaluation components/PSDA with CLOs					
SN	**Bloom's Level**	**Course Learning Outcomes**	**Class Test**	**Project**	**Viva**
1	**Understand**	Explain the fundamental concepts of data science and analytics, including the data science lifecycle and various types of analytics (descriptive, predictive and prescriptive).	✓	✓	✓
2	**Apply**	Use appropriate statistical methods and mathematical models to analyze and interpret data sets.	✓	✓	✓
3	**Analyze**	Categories the ability to collect, preprocess, clean, and manage structured and unstructured data using tools like Python, R, SQL, or Excel.	✓	✓	✓
4	**Create**	Create meaningful data visualizations to communicate findings clearly to stakeholders.	✓	✓	✓

Text Books:

1. Book James, G., Witten, D., Hastie, T., Tibshirani, R., "An introduction to statistical learning with app
2. Book Han, J., Kamber, M., Pei, J. "Data mining concepts and techniques", Morgan Kaufmann.
3. Book Hastie, T., Tibshirani, R., Friedman, J., " The Elements of Statistical Learning", Springer.

4. Book Murphy, K.,“Machine Learning: A Probabilistic Perspective” - MIT Press.
5. Book “Practical Data Science with R”. Nina Zumel, John Mount. Manning.
6. Book “Foundations of Statistical Natural Language Processing” by Christopher D.Manning and Hinrich Schuet
7. Book Steven Bird,Ewan Klein and Edward Loper “Natural Language Processing with Python”,O’Reilly Media.